# Surface Mount Schottky Power Rectifier

## SMA/SMB Power Surface Mount Package

# MBRS2H100T3G, NBRS2H100NT3G, MBRA2H100T3G, NRVBA2H100NT3G

This device employs the Schottky Barrier principle in a metal-to-silicon power rectifier. Features epitaxial construction with oxide passivation and metal overlay contact. Ideally suited for low voltage, high frequency switching power supplies; free wheeling diodes and polarity protection diodes.

### Features

- Compact Package with J-Bend Leads Ideal for Automated Handling
- Highly Stable Oxide Passivated Junction
- Guard-Ring for Overvoltage Protection
- Low Forward Voltage Drop
- NBR and NRVB Prefixes for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable*

### Mechanical Characteristics

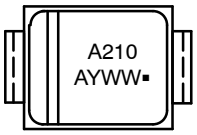
- Case: Molded Epoxy
- Epoxy Meets UL 94 V-0 @ 0.125 in
- Weight: 70 mg (SMA), 95 mg (SMB) (Approximately)
- Cathode Polarity Band
- Lead and Mounting Surface Temperature for Soldering Purposes: 260 °C Max. for 10 Seconds
- Finish: All External Surfaces Corrosion Resistant and Terminal Leads are Readily Solderable
- ESD Ratings:
  - Charged Device Model > 1000 V (Class C5)
  - Human Body Model = 3B
- These Devices are Pb-Free and are RoHS Compliant
- Device Meets MSL1 Requirements

## SCHOTTKY BARRIER RECTIFIER 2.0 AMPERES, 100 VOLTS

### MARKING DIAGRAMS

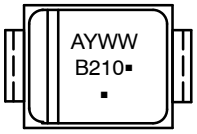


**SMA CASE 403D**






**SMB CASE 403A**




| | |
|---|---|
| A210 | = MBRA2H100T3G |
| B210 | = MBRS2H100T3G |
| A | = Assembly Location |
| Y | = Year |
| WW | = Work Week |
| ▪ | = Pb-Free Package |

(Note: Microdot may be in either location)

**The Assembly Location code (A) is front side optional. In cases where the Assembly Location is stamped in the package, the front side assembly code may be blank.

### ORDERING INFORMATION

| Device | Package | Shipping† |
|---|---|---|
| MBRA2H100T3G | SMA (Pb-Free) | 5,000 / Tape & Reel |
| MBRS2H100T3G, NBRS2H100NT3G* | SMB (Pb-Free) | 2,500 / Tape & Reel |
| NRVBA2H100NT3G* | SMA (Pb-Free) | 5,000 / Tape & Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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## MAXIMUM RATINGS

| Rating | Symbol | Value | Unit |
|---|---|---|---|
| Peak Repetitive Reverse Voltage<br>Working Peak Reverse Voltage<br>DC Blocking Voltage | $V_{RRM}$<br>$V_{RWM}$<br>$V_R$ | 100 | V |
| Average Rectified Forward Current<br>($T_L$ = 150 °C) | $I_O$ | 2.0 | A |
| Non-Repetitive Peak Surge Current<br>(Surge Applied at Rated Load Conditions Halfwave, Single Phase, 60 Hz) | $I_{FSM}$ | 130 | A |
| Storage Temperature Range | $T_{stg}$ | −65 to +175 | °C |
| Operating Junction Temperature (Note 1) | $T_J$ | −65 to +175 | °C |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. The heat generated must be less than the thermal conductivity from Junction-to-Ambient: $dP_D/dT_J < 1/R_{\theta JA}$.

## THERMAL CHARACTERISTICS

| Characteristic | Symbol | Value | Unit |
|---|---|---|---|
| Thermal Resistance, Junction-to-Lead (Note 2)<br>MBRA2H100T3G, NRVBA2H100NT3G<br>MBRS2H100T3G, NBRS2H100NT3G | $\Psi_{JCL}$ | <br>14<br>12 | °C/W |
| Thermal Resistance, Junction-to-Ambient (Note 2)<br>MBRA2H100T3G, NRVBA2H100NT3G<br>MBRS2H100T3G, NBRS2H100NT3G | $R_{\theta JA}$ | <br>75<br>71 | °C/W |
| Thermal Resistance, Junction-to-Ambient (Note 3)<br>MBRA2H100T3G, NRVBA2H100NT3G<br>MBRS2H100T3G, NBRS2H100NT3G | $R_{\theta JA}$ | <br>275<br>230 | °C/W |

2. Mounted with 700 mm square copper pad size (Approximately 1 inch square) 1 oz FR4 Board.
3. Mounted with minimum recommended pad size 1 oz FR4 Board.

## ELECTRICAL CHARACTERISTICS

| Characteristic | Symbol | Value | | Unit |
|---|---|---|---|---|
| | | $T_J$ = 25 °C | $T_J$ = 125 °C | |
| Maximum Instantaneous Forward Voltage (Note 4)<br>($i_F$ = 2.0 A) | $v_F$ | 0.79 | 0.65 | V |
| Maximum Instantaneous Reverse Current (Note 4)<br>($V_R$ = 100 V) | $I_R$ | 0.008 | 1.5 | mA |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

4. Pulse Test: Pulse Width ≤ 380 μs, Duty Cycle ≤ 2.0%.



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## TYPICAL CHARACTERISTICS

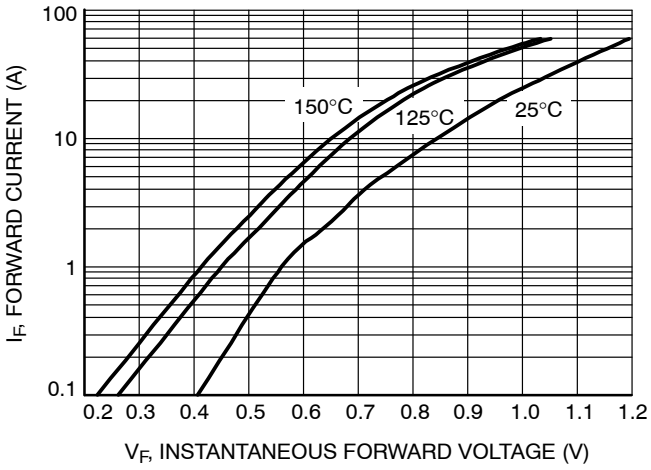



**Figure 1. Typical Forward Voltage**

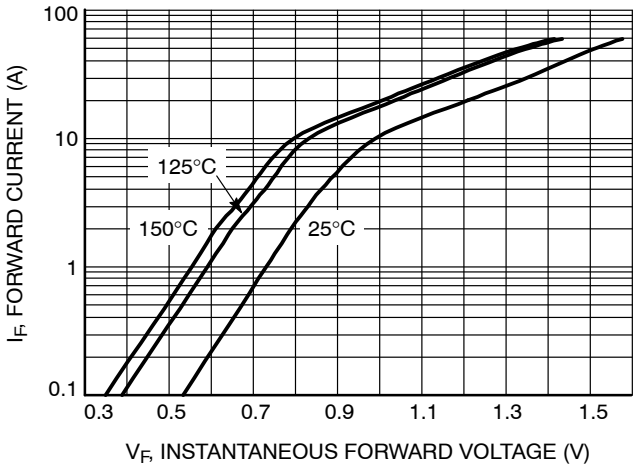



**Figure 2. Maximum Forward Voltage**

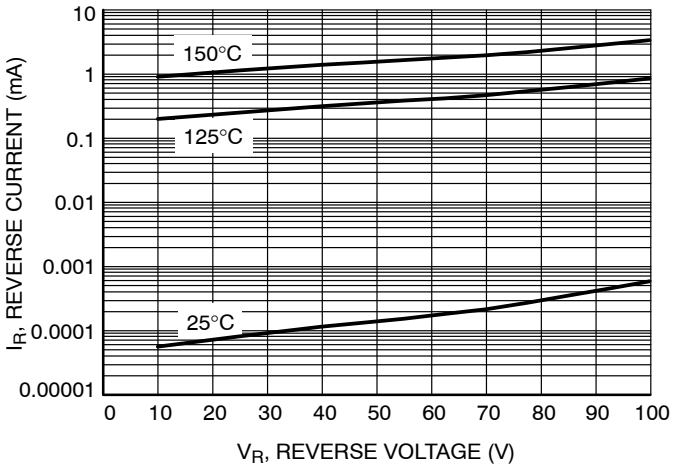



**Figure 3. Typical Reverse Current**

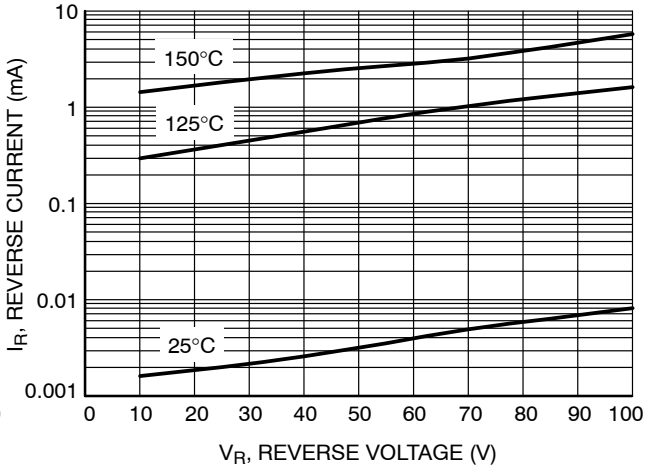



**Figure 4. Maximum Reverse Current**

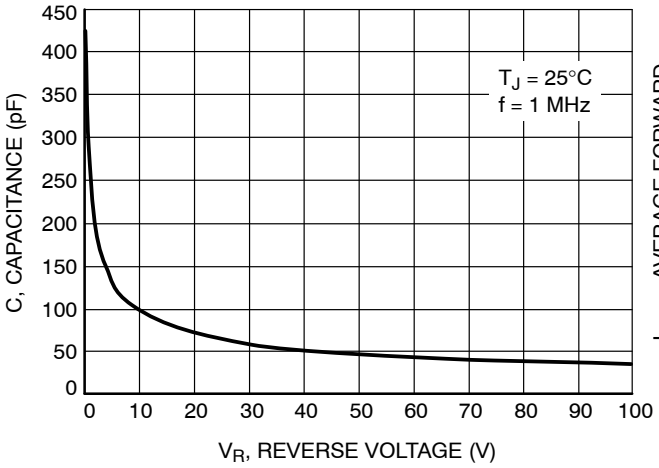



**Figure 5. Typical Capacitance**

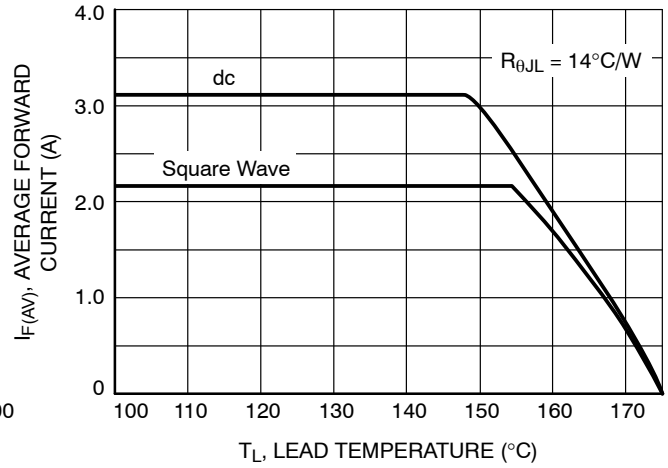



**Figure 6. Current Derating – Lead**



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## TYPICAL CHARACTERISTICS

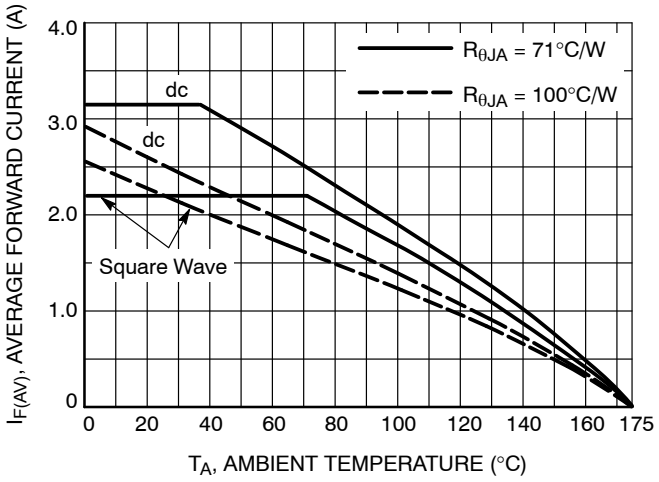



**Figure 7. Current Derating, Ambient**

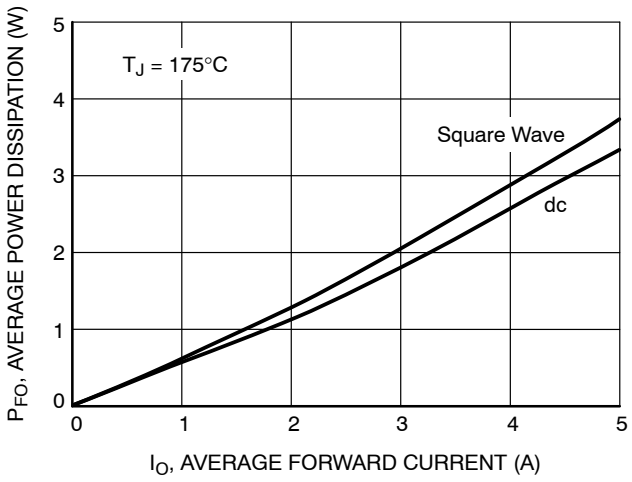



**Figure 8. Maximum Forward Power Dissipation**

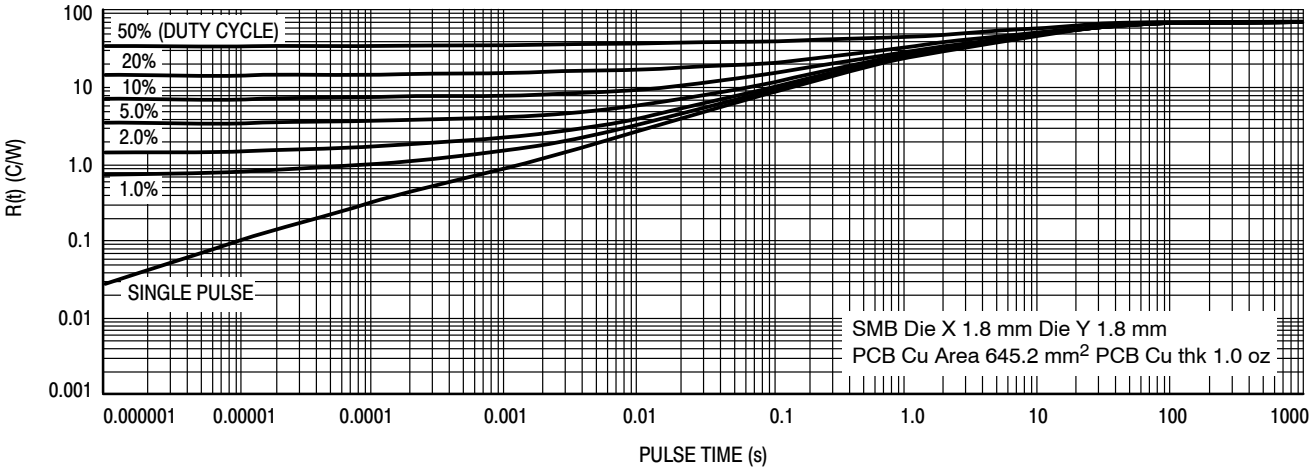



**Figure 9. Thermal Response, Junction-to-Ambient (1 inch pad) – MBRS2H100T3G/NBRS2H100NT3G**

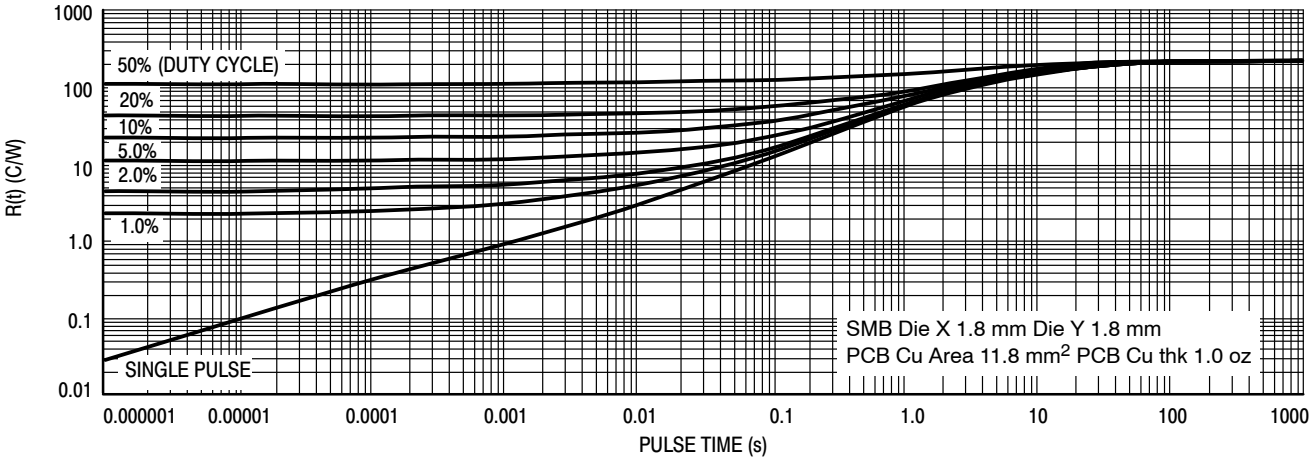



**Figure 10. Thermal Response, Junction-to-Ambient (min pad) – MBRS2H100T3G/NBRS2H100NT3G**



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## TYPICAL CHARACTERISTICS

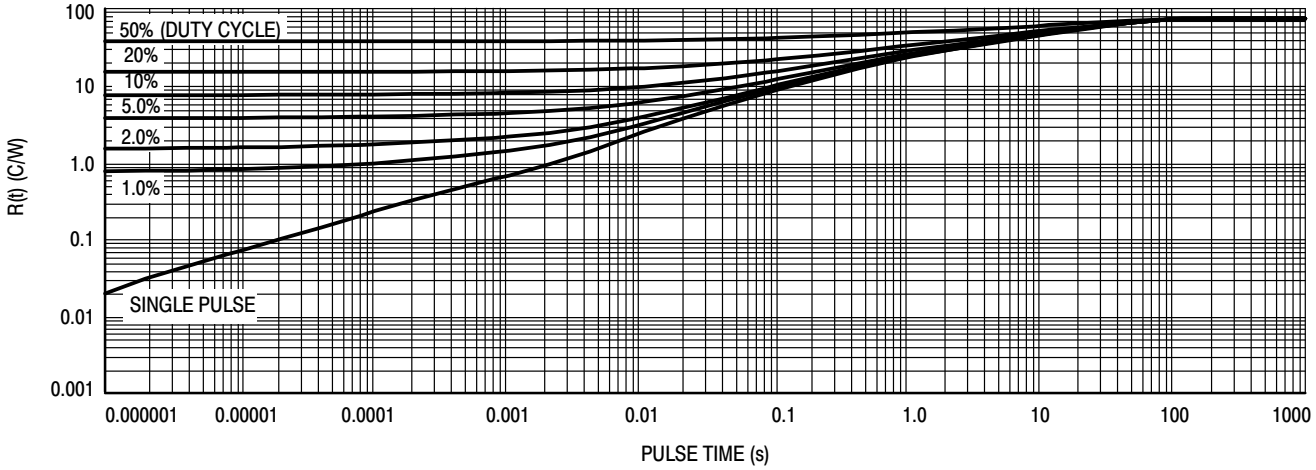



**Figure 11. Thermal Response, Junction-to-Ambient (1 inch pad) – MBRA2H100T3G**

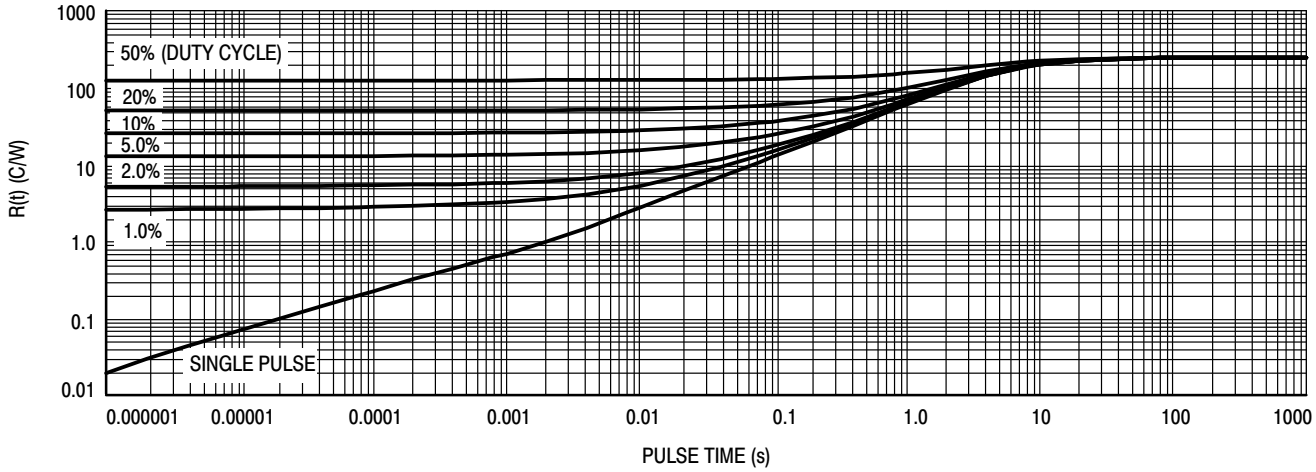



**Figure 12. Thermal Response, Junction-to-Ambient (min pad) – MBRA2H100T3G**

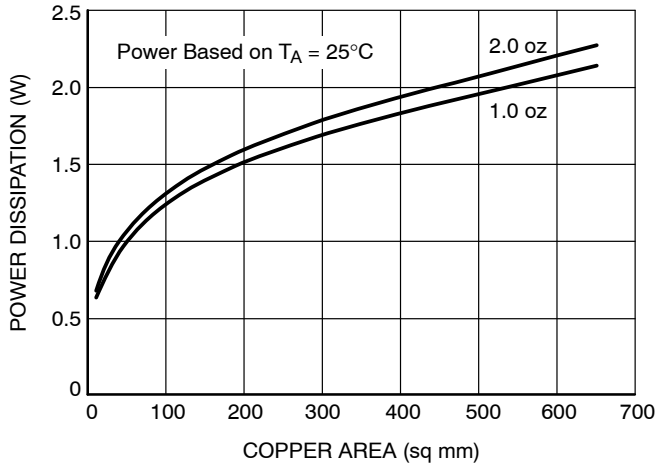



**Figure 13. $P_D$, Junction-to-Ambient (URS copper area)**



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## REVISION HISTORY

| Revision | Description of Changes | Date |
|---|---|---|
| 16 | Remove EOL device information | 8/1/2025 |

This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.



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**SMB**
CASE 403A−03
ISSUE J

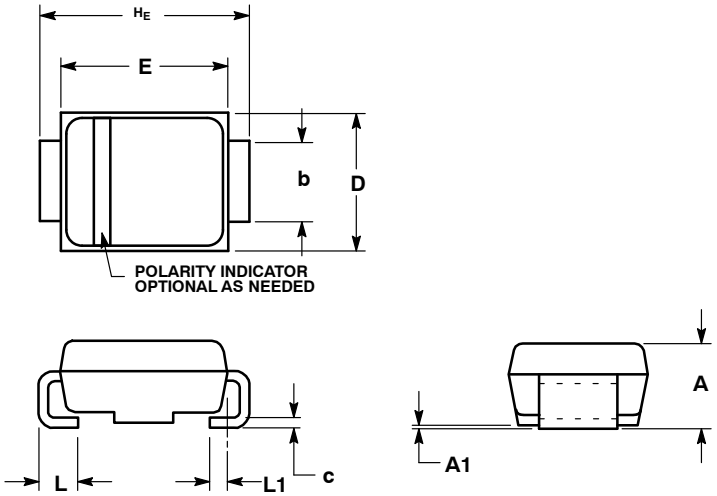
DATE 19 JUL 2012



**SCALE 1:1**
**Polarity Band**



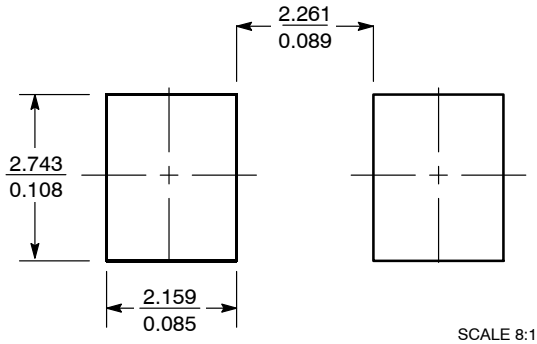
**SCALE 1:1**
**Non−Polarity Band**




NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH.
3. DIMENSION b SHALL BE MEASURED WITHIN DIMENSION L1.

| | MILLIMETERS | | | INCHES | | |
|---|---|---|---|---|---|---|
| DIM | MIN | NOM | MAX | MIN | NOM | MAX |
| A | 1.95 | 2.30 | 2.47 | 0.077 | 0.091 | 0.097 |
| A1 | 0.05 | 0.10 | 0.20 | 0.002 | 0.004 | 0.008 |
| b | 1.96 | 2.03 | 2.20 | 0.077 | 0.080 | 0.087 |
| c | 0.15 | 0.23 | 0.31 | 0.006 | 0.009 | 0.012 |
| D | 3.30 | 3.56 | 3.95 | 0.130 | 0.140 | 0.156 |
| E | 4.06 | 4.32 | 4.60 | 0.160 | 0.170 | 0.181 |
| $H_E$ | 5.21 | 5.44 | 5.60 | 0.205 | 0.214 | 0.220 |
| L | 0.76 | 1.02 | 1.60 | 0.030 | 0.040 | 0.063 |
| L1 | 0.51 REF | | | 0.020 REF | | |

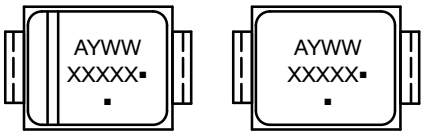
## SOLDERING FOOTPRINT*




SCALE 8:1 (mm / inches)

*For additional information on our Pb−Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

## GENERIC MARKING DIAGRAM*




**Polarity Band** **Non−Polarity Band**

XXXXX = Specific Device Code
A = Assembly Location
Y = Year
WW = Work Week
▪ = Pb−Free Package

(Note: Microdot may be in either location)

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

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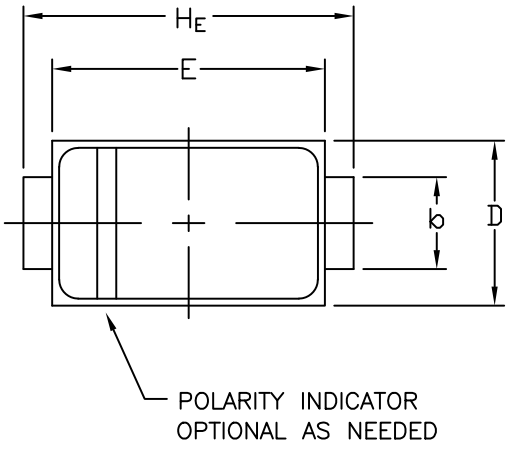
**SMA**
CASE 403D
ISSUE J

DATE 22 OCT 2021



STYLE 1 STYLE 2

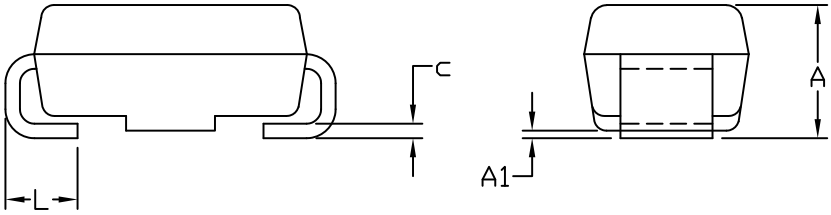
**SCALE 1:1**




NOTES:

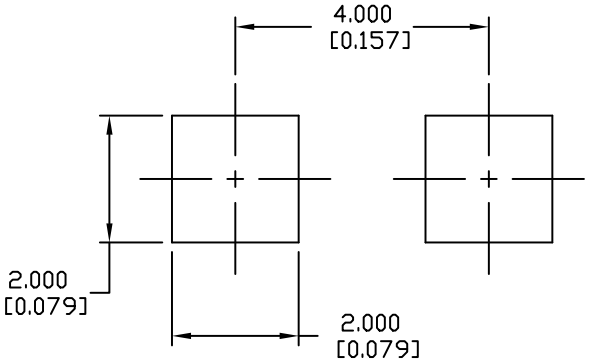
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCHES
3. DIMENSION b SHALL BE MEASURED WITHIN DIMENSION L.

| | MILLIMETERS | | | INCHES | | |
|---|---|---|---|---|---|---|
| DIM | MIN. | NOM. | MAX. | MIN. | NOM. | MAX. |
| A | 1.97 | 2.10 | 2.20 | 0.078 | 0.083 | 0.087 |
| A1 | 0.05 | 0.10 | 0.20 | 0.002 | 0.004 | 0.008 |
| b | 1.27 | 1.45 | 1.63 | 0.050 | 0.057 | 0.064 |
| c | 0.15 | 0.28 | 0.41 | 0.006 | 0.011 | 0.016 |
| D | 2.29 | 2.60 | 2.92 | 0.090 | 0.103 | 0.115 |
| E | 4.06 | 4.32 | 4.57 | 0.160 | 0.170 | 0.180 |
| HE | 4.83 | 5.21 | 5.59 | 0.190 | 0.205 | 0.220 |
| L | 0.76 | 1.14 | 1.52 | 0.030 | 0.045 | 0.060 |



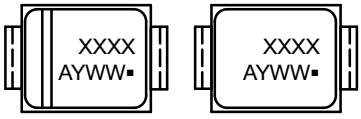

STYLE 1:
PIN 1. CATHODE (POLARITY BAND)
2. ANODE

STYLE 2:
NO POLARITY




RECOMMENDED
MOUNTING FOOTPRINT

## GENERIC MARKING DIAGRAM*




STYLE 1 STYLE 2

XXXX = Specific Device Code
A = Assembly Location
Y = Year
WW = Work Week
▪ = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

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